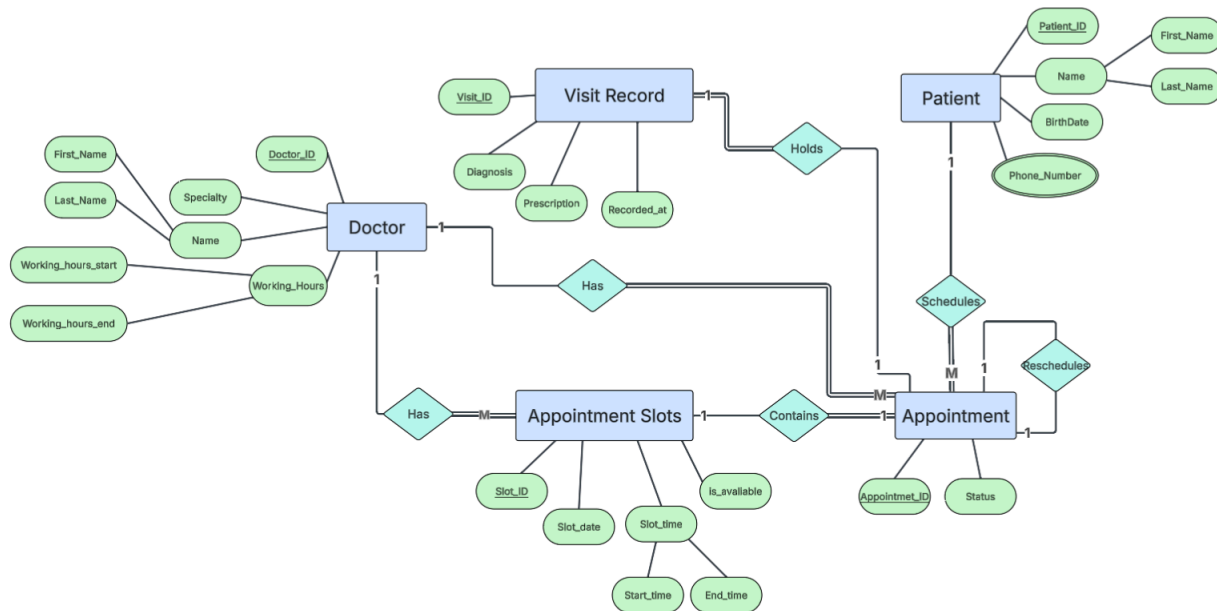
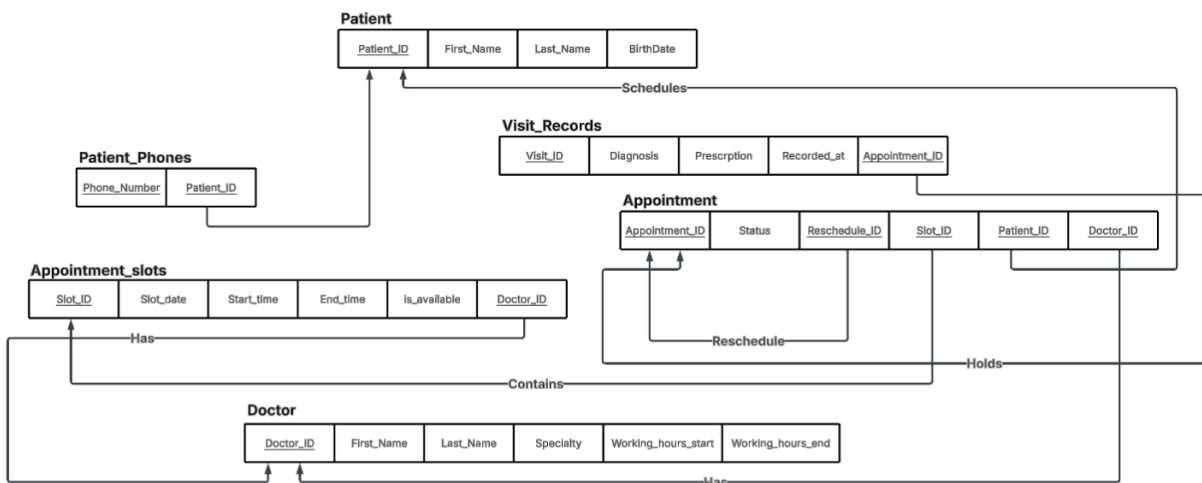


Clinic Appointment & Patient Records API

ERD & Mapping



Activate Windows
Go to Settings to activate Windows.



To solve the cancellation-without-deletion problem I created an entity called appointment where all appointments are recorded. A newly created appointment is labelled as a scheduled appointment if that appointment is rescheduled then it's assigned a reschedule id of the old appointment and a new appointment id, cancelled appointments are just labelled cancelled and a new appointment should be made if the user wishes to do so. Appointment_slots entity contains all the active (scheduled) appointments. So Appointment entity acts as a history for all appointment statuses while appointment slot is only a history for scheduled appointments.

Testing

Creating a patient:

The screenshot shows a REST client interface with the following components:

- Header:** [CONFLICT] POST Get da, No environment.
- Toolbar:** My Collection > Get data, Save, Share.
- Request Section:**
 - Method:** POST
 - URL:** http://localhost:8080/patients
 - Body:** raw (selected), JSON (selected). The body content is:

```
1 {
2   "firstName": "Khadija",
3   "lastName": "Al-Mamari",
4   "birthDate": "2000-03-15",
5   "phones": ["98772211", "79112034"]
6 }
```
 - Buttons:** Schema, Beautify.
- Response Section:**
 - Status:** 201 Created
 - Time:** 194 ms
 - Size:** 281 B
 - Body:** JSON (selected). The response content is:

```
1 {
2   "id": 21,
3   "firstName": "Khadija",
4   "lastName": "Al-Mamari",
5   "birthDate": "2000-03-15",
6   "phones": [
7     "98772211",
8     "79112034"
9   ]
10 }
```

Getting a specific patient:

The screenshot shows an HTTP client interface with the following details:

- Collection:** My Collection > Get data
- Method:** GET
- URL:** http://localhost:8080/patients/1
- Buttons:** Save, Share, Send
- Body:** raw (selected), JSON, Schema, Beautify
- Response:** 200 OK, 52 ms, 280 B
- JSON Data:**

```
1 {
2   "id": 1,
3   "firstName": "Ahmed",
4   "lastName": "Al-Balushi",
5   "birthDate": "1990-03-15",
6   "phones": [
7     "96812345678",
8     "96887654321"
9   ]
10 }
```

Creating a doctor:

The screenshot displays a REST client interface with the following components:

- Header:** Shows a status bar with "[CONFLICT] POST Get da", a dropdown menu, and "No environment".
- Toolbar:** Includes "My Collection > Get data", "Save", and "Share" buttons.
- Request Configuration:** A dropdown menu set to "POST" and a text field containing the URL "http://localhost:8080/doctors". A "Send" button is to the right.
- Body Editor:** Labeled "Body" with a dropdown menu. It shows the request body in "JSON" format, with options for "raw", "Schema", and "Beautify". The JSON content is:

```
1 {
2   "firstName": "Khalid",
3   "lastName": "Al-Salmi",
4   "speciality": "Ortho",
5   "workingHoursStart": "10:00",
6   "workingHoursEnd": "19:00"
7 }
```
- Response Section:** Labeled "Body" with a refresh icon. It shows a green status bar indicating "201 Created" with a response time of "22 ms" and a size of "302 B". Below this, the response body is shown in "JSON" format:

```
1 {
2   "id": 24,
3   "firstName": "Khalid",
4   "lastName": "Al-Salmi",
5   "speciality": "Ortho",
6   "workingHoursStart": "10:00:00",
7   "workingHoursEnd": "19:00:00"
8 }
```

Getting a specific doctor:

The screenshot shows a REST client interface with the following components:

- Header:** "My Collection > Get data" with "Save" and "Share" buttons.
- Request:** Method "GET" and URL "http://localhost:8080/doctors/2" with a "Send" button.
- Body:** A large empty text area with a "1" at the top left.
- Response:** Status "200 OK", time "74 ms", size "315 B", and a "JSON" tab selected.
- JSON Response:**

```
1 {
2   "id": 2,
3   "firstName": "Mohammed",
4   "lastName": "Al-Balushi",
5   "speciality": "Ear, Nose and Throat",
6   "workingHoursStart": "10:00:00",
7   "workingHoursEnd": "20:00:00"
8 }
```

Doctor that does not exist yet:

The screenshot shows a REST client interface with the following components:

- Header:** [CONFLICT] GET Get dat + No environment
- Toolbar:** HTTP icon, My Collection > Get data, Save, Share, New Chat
- Request Section:**
 - Method: GET
 - URL: http://localhost:8080/doctors/3/sched ...
 - Send button
 - Body dropdown
 - Body content: 1
 - JSON dropdown
 - Schema and Beautify buttons
- Response Section:**
 - Status: 404 Not Found (highlighted in red)
 - Details: 11 ms • 245 B • [Globe icon] | e.g. ...
 - JSON dropdown
 - Response body (formatted JSON):

```
1 {
2   "status": 404,
3   "error": "Not Found",
4   "message": "Doctor not found with ID: 3"
5 }
```

Patient that does not exist yet:

The screenshot shows a REST client interface with the following components:

- Header:** "My Collection > Get data" with "Save" and "Share" buttons.
- Request Section:** Method "GET" and URL "http://localhost:8080/patients/31" with a "Send" button.
- Body Section:** A large empty text area with a "raw" dropdown and "JSON" selected. "Schema" and "Beautify" buttons are also present.
- Response Section:** Shows a "404 Not Found" status in a red box, with "8 ms" and "247 B" indicating response time and size. The response body is displayed in a JSON format:

```
1 {
2   "status": 404,
3   "error": "Not Found",
4   "message": "Patient not found with ID: 31"
5 }
```

Doctor with free schedule:

 My Collection > **Get data**  Save  Share

GET 

http://localhost:8080/doctors/23/schei...

Send 

Body  

raw 

JSON 

 Schema 

1

Body  

200 OK • 10 ms • 166 B •  |  e.g. ... 

 JSON 

1

[]

Doctor schedule with patients:

The screenshot shows a REST client interface with the following details:

- URL Bar:** Method: GET, URL: `http://localhost:8080/doctors/23/sche...`, Buttons: Save, Share, Send.
- Body Tab:** raw (selected), JSON, Schema, Beautify. The raw view shows the number "1".
- Status Bar:** 200 OK, 18 ms, 422 B.
- Response View:** JSON (selected). The response is a JSON array with 2 elements:

```
1  [  
2    {  
3      "slotId": 42,  
4      "startTime": "09:00:00",  
5      "endTime": "09:30:00",  
6      "available": false,  
7      "appointmentId": 22,  
8      "patientName": "Khadija Al-Mamari"  
9    },  
10   {  
11     "slotId": 43,  
12     "startTime": "13:00:00",  
13     "endTime": "13:30:00",  
14     "available": false,  
15     "appointmentId": 23,  
16     "patientName": "Salwa Al-Busaidi"  
17   }  
18 ]
```

Patient previous visits:

The screenshot shows a REST client interface with a GET request to `http://localhost:8080/patients/1/history`. The response is a 200 OK status with a JSON body containing two visit records.

Request:

- Method: GET
- URL: `http://localhost:8080/patients/1/history`

Response:

Status: 200 OK • 433 ms • 1.97 KB

Body (JSON):

```
1  [
2    {
3      "appointmentId": 6,
4      "status": "SCHEDULED",
5      "appointmentDate": "2026-07-15",
6      "startTime": "10:30:00",
7      "endTime": "11:00:00",
8      "doctorName": "Mohammed Al-Balushi",
9      "doctorSpeciality": "Ear, Nose and Throat",
10     "rescheduledToAppointmentId": null,
11     "visitId": null,
12     "diagnosis": null,
13     "prescription": null,
14     "recordedAt": null
15   },
16   {
17     "appointmentId": 1,
18     "status": "CANCELLED",
19     "appointmentDate": "2026-07-10",
20     "startTime": "10:00:00",
21     "endTime": "10:30:00",
22     "doctorName": "Sarah Al-Rashdi",
23     "doctorSpeciality": "Cardiology",
24     "rescheduledToAppointmentId": null,
25     "visitId": null,
26     "diagnosis": null,
27     "prescription": null,
28     "recordedAt": null
29   }
30 ]
```

Patient with no previous visit history:

The screenshot shows a REST client interface with the following components:

- Header:** "My Collection > Get data" with "Save" and "Share" buttons.
- Request Bar:** Method "GET" and URL "http://localhost:8080/patients/2/history". A "Send" button is on the right.
- Body Tab:** Labeled "Body" with a dropdown arrow. It contains a "raw" tab and a "JSON" tab. To the right are "Schema" and "Beautify" links.
- Response Area:** A large text area containing the number "1".
- Status Bar:** Shows "200 OK", "10 ms", "166 B", and a "e.g." link.
- Response Tab:** Labeled "{} JSON" with a dropdown arrow. It shows the response "1" followed by an empty array "[]".

Booking an appointment:

[CONFLICT] POST Get da +

No environment

My Collection > Get data Save Share

POST http://localhost:8080/appointments Send

Body Cookies

raw JSON Schema Beautify

```
1 {
2   "patientId": 21,
3   "doctorId": 23,
4   "date": "2026-07-10",
5   "startTime": "09:00"
6 }
```

Body 201 Created • 58 ms • 407 B • e.g. ...

{ } JSON

```
1 {
2   "appointmentId": 22,
3   "status": "SCHEDULED",
4   "patientId": 21,
5   "patientName": "Khadija Al-Mamari",
6   "doctorId": 23,
7   "doctorName": "Ahmed Al-Hamdani",
8   "slotDate": "2026-07-10",
9   "startTime": "09:00:00",
10  "endTime": "09:30:00",
11  "rescheduledToAppointmentId": null
12 }
```

Cancelling an appointment:

The screenshot shows a REST client interface with the following components:

- Header:** "My Collection > Get data" with "Save" and "Share" buttons.
- Method and URL:** A dropdown menu set to "POST" and a text field containing "http://localhost:8080/appointments/8/...". A "Send" button is to the right.
- Body:** A section labeled "Body" with a "raw" dropdown and a "JSON" dropdown. Below these is a large empty text area with a "1" in the top left corner.
- Response:** A status bar shows "200 OK" in green, followed by "85 ms", "401 B", and a globe icon. Below this is a "JSON" dropdown and a play button. The response body is a JSON object displayed with line numbers 1 through 12.

```
1  {
2    "appointmentId": 8,
3    "status": "CANCELLED",
4    "patientId": 2,
5    "patientName": "Salwa Al-Busaidi",
6    "doctorId": 2,
7    "doctorName": "Mohammed Al-Balushi",
8    "slotDate": "2026-07-15",
9    "startTime": "10:35:00",
10   "endTime": "11:05:00",
11   "rescheduledToAppointmentId": null
12 }
```

Cancelling an appointment that isnt scheduled:

The screenshot shows a REST client interface with the following elements:

- Header:** [CONFLICT] POST Get da, No environment, Save, Share.
- Method and URL:** POST, http://localhost:8080/appointments/1/c...
- Body:** raw, JSON, Schema, Beautify. The body content is "1".
- Response:** 400 Bad Request, 13 ms, 264 B. The response body is a JSON object:

```
{  "status": 400,  "error": "Bad Request",  "message": "Only SCHEDULED appointments can be cancelled. Current status: CANCELLED"}
```

Rescheduling an appointment:

The screenshot shows a REST client interface with the following components:

- Top Bar:** Includes a "My Collection > Get data" breadcrumb, "Save" and "Share" buttons, and a dropdown menu.
- Request Section:**
 - Method:** POST
 - URL:** http://localhost:8080/appointments/23...
 - Buttons:** "Send" and a dropdown menu.
 - Body:** A dropdown menu showing "raw" and "JSON" options.
 - Schema:** A button labeled "Schema" with a link icon.
 - Beautify:** A button labeled "Beautify".
- Request Body (JSON):**

```
1 {
2   "newDate": "2026-07-15",
3   "newStartTime": "11:00"
4 }
```
- Response Section:**
 - Status:** 200 OK
 - Time:** 44 ms
 - Size:** 400 B
 - Body:** A dropdown menu showing "{} JSON" and icons for play, image, and dropdown.
 - Response Body (JSON):**

```
1 {
2   "appointmentId": 24,
3   "status": "SCHEDULED",
4   "patientId": 2,
5   "patientName": "Salwa Al-Busaidi",
6   "doctorId": 23,
7   "doctorName": "Ahmed Al-Hamdani",
8   "slotDate": "2026-07-15",
9   "startTime": "11:00:00",
10  "endTime": "11:30:00",
11  "rescheduledToAppointmentId": null
12 }
```

Rescheduling an appointment that isn't scheduled:

The screenshot shows a REST client interface with the following details:

- Header:** [CONFLICT] POST Get da, No environment.
- Toolbar:** My Collection > Get data, Save, Share.
- Request:** Method: POST, URL: http://localhost:8080/appointments/1/r..., Send button.
- Body:** raw, JSON, Schema, Beautify. The body content is a single line: 1.
- Response:** Body, 500 Internal Server Error, 5 ms, 241 B. The response is displayed in JSON format:

```
1 {
2   "status": 500,
3   "error": "Internal Server Error",
4   "message": "An unexpected error occurred"
5 }
```


Completing an appointment:

The screenshot shows a REST client interface with the following components:

- Top Bar:** Includes an HTTP icon, "My Collection > Get data", a "Save" button, and a "Share" button with a link icon.
- Request Section:** A dropdown menu shows "POST", and the URL field contains "http://localhost:8080/appointments/21 ...". A blue "Send" button is to the right.
- Body Section:** Labeled "Body" with a dropdown arrow. It contains a "raw" dropdown and a "JSON" dropdown. To the right are "Schema" and "Beautify" links.
- Response Section:** A large text area displays the number "1".
- Status Bar:** Shows "Body" with a refresh icon, a green status "200 OK", and response details: "27 ms", "402 B", a globe icon, "e.g.", and a three-dot menu.
- JSON View:** A dropdown shows "{} JSON" with a play icon and a screenshot icon. To the right are icons for expand/collapse, search, and copy/link.
- JSON Data:** A pre-formatted JSON object is displayed with line numbers 1 through 12:

```
1 {
2   "appointmentId": 21,
3   "status": "COMPLETED",
4   "patientId": 2,
5   "patientName": "Salwa Al-Busaidi",
6   "doctorId": 2,
7   "doctorName": "Mohammed Al-Balushi",
8   "slotDate": "2026-07-10",
9   "startTime": "10:00:00",
10  "endTime": "10:30:00",
11  "rescheduledToAppointmentId": null
12 }
```

Completing an appointment that isn't scheduled:

The screenshot shows a REST client interface with the following components:

- Header:** "My Collection > Get data" with "Save" and "Share" buttons.
- Method and URL:** A dropdown menu set to "POST" and a text field containing "http://localhost:8080/appointments/1/c...". A "Send" button is to the right.
- Body Tab:** Labeled "Body" with a dropdown menu. It contains a "raw" tab and a "JSON" tab. To the right are "Schema" and "Beautify" links.
- Request Body:** A large text area containing the number "1".
- Response Section:** Below the request body, it shows "Body" with a refresh icon, a status bar indicating "400 Bad Request", and performance metrics: "41 ms", "274 B", and a globe icon.
- Response Body:** A JSON object displayed in a code editor with line numbers 1 through 5. The JSON is:

```
{  "status": 400,  "error": "Bad Request",  "message": "Only SCHEDULED appointments can be marked as completed. Current status: CANCELLED"}
```

Recording a visit:

 My Collection > Get data  Save  Share

POST  http://localhost:8080/appointments/21 ...  Send 

Body  Cookies

raw  JSON  Schema Beautify

```
1  {
2    "diagnosis": "Hypertension Stage 1",
3    "prescription": "Panadol 20mg once daily for 30
4    days"
}
```

Body  201 Created • 64 ms • 405 B •  e.g. ...

{ } JSON         

```
1  {
2    "visitId": 21,
3    "appointmentId": 21,
4    "patientId": 2,
5    "patientName": "Salwa Al-Busaidi",
6    "visitDate": "2026-07-10",
7    "diagnosis": "Hypertension Stage 1",
8    "prescription": "Panadol 20mg once daily for 30
9    days",
10   "recordedAt": "2026-06-23T14:42:21.9510179"
}
```

Trying to do visit for not a completed appointment:

The screenshot shows a REST client interface with the following details:

- Header:** [CONFLICT] POST Get da, No environment.
- Toolbar:** HTTP icon, My Collection > Get data, Save, Share.
- Request:** Method: POST, URL: http://localhost:8080/appointments/23..., Send button.
- Body:** Tabbed view with 'raw' and 'JSON' options. The 'JSON' tab is active, showing a valid JSON object:

```
1 {
2   "diagnosis": "Hypertension Stage 1",
3   "prescription": "Panadol 20mg once daily for 30
4   days"
}
```
- Response:** Status: 400 Bad Request, 18 ms, 275 B. The response body is shown in a JSON tab:

```
1 {
2   "status": 400,
3   "error": "Bad Request",
4   "message": "Visit can only be recorded for
5   COMPLETED appointments. Current status:
   RESCHEDULED"
}
```

Business Logic

Working hours of a doctor must be before working hours end:

The screenshot shows an HTTP client interface. At the top, it says "My Collection > Get data" with "Save" and "Share" buttons. Below this, a dropdown menu shows "POST" and the URL "http://localhost:8080/doctors". A "Send" button is to the right. Under "Body", there are tabs for "raw" and "JSON", with "JSON" selected. The JSON body is:

```
{  "firstName": "Sarah",  "lastName": "Al-Rashdi",  "speciality": "Cardiology",  "workingHoursStart": "16:00",  "workingHoursEnd": "08:00"}
```

 Below the body, a status bar shows "400 Bad Request" in a red box, followed by "7 ms", "245 B", and a globe icon. Underneath, there's a "JSON" tab and a play button. The response body is:

```
{  "status": 400,  "error": "Bad Request",  "message": "Working hours start must be before working hours end"}
```

My Collection > Get data Save Share

POST http://localhost:8080/doctors Send

Body Cookies

raw JSON Schema Beautify

```
1 {
2   "firstName": "Sarah",
3   "lastName": "Al-Rashdi",
4   "speciality": "Cardiology",
5   "workingHoursStart": "16:00",
6   "workingHoursEnd": "08:00"
7 }
```

Body 400 Bad Request 7 ms 245 B e.g. ...

{ } JSON { } { } { } { }

```
1 {
2   "status": 400,
3   "error": "Bad Request",
4   "message": "Working hours start must be before
5             working hours end"
6 }
```

Appointment date must not be in the past of the current time and date:

[CONFLICT] POST Get da + No environment

My Collection > Get data Save Share

POST http://localhost:8080/appointments Send

Body Cookies

raw JSON Schema Beautify

```
1 {
2   "patientId": 22,
3   "doctorId": 21,
4   "date": "2026-06-10",
5   "startTime": "09:00"
6 }
```

Body 400 Bad Request • 22 ms • 245 B • e.g. ...

{ } JSON

```
1 {
2   "status": 400,
3   "error": "Validation Failed",
4   "message": "date: Appointment date must not be
5             in the past"
```

An appointment shouldnt be booked outside of doctors working hours:

The screenshot shows a REST client interface with a tab labeled "[CONFLICT] POST Get da". The request is a POST to "http://localhost:8080/appointments". The request body is a JSON object:

```
1 {
2   "patientId": 21,
3   "doctorId": 21,
4   "date": "2026-07-10",
5   "startTime": "23:00"
6 }
```

The response is a "400 Bad Request" with a status of 400, a duration of 24 ms, and a size of 267 B. The response body is a JSON object:

```
1 {
2   "status": 400,
3   "error": "Bad Request",
4   "message": "Requested time 23:00 is outside the
5             doctor's working hours (10:00 - 18:00)"
6 }
```

Patient cannot have two active appointments with the same doctor on the same day:

The screenshot shows an HTTP client interface. At the top, there's a navigation bar with "My Collection > Get data", a "Save" button, and a "Share" button. Below this, the request method is set to "POST" and the URL is "http://localhost:8080/appointments". A "Send" button is visible. The "Body" tab is selected, showing a JSON payload:

```
1 {
2   "patientId": 21,
3   "doctorId": 22,
4   "date": "2026-07-10",
5   "startTime": "11:00"
6 }
```

Below the request, the response status is "409 Conflict" with a duration of "23 ms" and a size of "308 B". The response body is shown in JSON format:

```
1 {
2   "status": 409,
3   "error": "Conflict",
4   "message": "Patient already has an appointment
5             with another doctor between 11:00 and 11:30
6             on 2026-07-10"
7 }
```


Patient cannot be booked with any doctor in an overlapping time range with another appointment:

The screenshot shows a REST client interface with the following details:

- Request:**
 - Method: **POST**
 - URL: `http://localhost:8080/appointments`
 - Body (JSON):

```
1 {
2   "patientId": 1,
3   "doctorId": 22,
4   "date": "2026-07-15",
5   "startTime": "11:00"
6 }
```
- Response:**
 - Status: **409 Conflict** (18 ms, 278 B)
 - Body (JSON):

```
1 {
2   "status": 409,
3   "error": "Conflict",
4   "message": "Doctor is already booked between
               11:00 and 11:30 on 2026-07-15"
5 }
```

Cannot record two visits for the same appointment :

The screenshot shows a REST client interface with the following details:

- Request:**
 - Method: **POST**
 - URL: `http://localhost:8080/appointments/21...`
 - Body (JSON):

```
1 {
2   "diagnosis": "Hypertension Stage 1",
3   "prescription": "Amlodipine 5mg once daily for
4     30 days"
```
- Response:**
 - Status: **409 Conflict** (14 ms, 268 B)
 - Body (JSON):

```
1 {
2   "status": 409,
3   "error": "Conflict",
4   "message": "A visit record already exists for
5     appointment ID: 21"
```

patients are not able to book or reschedule in between a pre-existing appointment time slot:

The screenshot displays a REST client interface with the following components:

- Header:** Shows a status of **[CONFLICT]** for a **POST** request to **Get da**. The environment is set to **No environment**.
- Toolbar:** Includes **Save** and **Share** buttons.
- Request Section:**
 - Method:** **POST**
 - URL:** **http://localhost:8080/appointments**
 - Body:** A JSON object:

```
{  "patientId": 2,  "doctorId": 22,  "date": "2026-07-15",  "startTime": "11:10"}
```
 - Buttons:** **Send** and **Schema** (disabled).
- Response Section:**
 - Status:** **409 Conflict** (17 ms, 278 B)
 - Body:** A JSON object:

```
{  "status": 409,  "error": "Conflict",  "message": "Doctor is already booked between 11:10 and 11:40 on 2026-07-15"}
```